

Weakly supervised learning for snow cover segmentation in mountainous areas from Sentinel-1 SAR images using interpolated NDSI time series

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ARTICLE INFO

Dataset link: <https://doi.org/10.57745/IMTSFL>

Keywords:

Snow cover
Optical
SAR
Time series
Pseudo-labels
Weak supervision
Kalman smoother

ABSTRACT

Snow cover plays a fundamental role in climate regulation and hydrological processes. Existing snow products are mainly based on optical imagery. Yet, snow monitoring remains challenging in mountainous regions due to frequent cloud cover. Synthetic Aperture Radar (SAR) imagery, unaffected by clouds, enables regular wet snow observations. However dry snow remains mostly transparent to SAR. In this study, we propose a fully automated framework that uses optical-derived labels to train a SAR-based model, combining the ability of optical sensors to detect snow with the cloud-penetrating capability of SAR. At inference time, our method only requires SAR images to predict snow maps, encompassing wet and dry snow. A convolutional neural network is trained to predict a binary snow cover map from a Sentinel-1 Single Look Complex (SLC) dual-pol amplitude image and a snow-free reference image. We generate binary training labels from thresholded MODIS Normalized Difference Snow Index (NDSI). Our model is trained in a weakly supervised manner by filling the cloud-induced gaps via temporal interpolation to generate pseudo-labels from sparse observations. We first evaluate the influence of the input SAR channels configuration and show that concatenating the acquisition of the day with the reference image is preferable to more complex preprocessing. Then, we compare the Closest Neighbours Interpolation and the Kalman smoother to fill the cloud-induced gaps in the MODIS NDSI time series. We show that increasing the level of supervision improves the model performance. By removing all the gaps and the noise in the NDSI time series, the Kalman smoother yields the best model performance. However the regularization strength of the Kalman smoother is shown to be critical. To validate our method, we compare it to existing snow products. By comparing with the THEIA L2B Snow product, we show that our method gives comparable results to Sentinel-2 based snow cover maps. The comparison with the Copernicus Wet/Dry Snow product shows that our model can detect both wet and dry snow solely from Sentinel-1 dual-pol amplitude images. Overall, this study highlights how the snow detection capabilities of optical sensor can be transferred to SAR images thanks to a deep learning framework, to ensure a robust monitoring of snow in cloud-prone alpine regions.

1. Introduction

Earth's cryosphere is one of the main drivers of its climate variability, due to its high albedo (0.8 to 0.9 for fresh snow). Indeed, snow covers up to 49% of the Northern Hemisphere land surface during midwinter time and it stores 75% of fresh water on land (Lemke et al., 2007). Mountain ranges are considered as the “water-towers of the world” (Viviroli et al., 2007), providing freshwater for the adjacent areas downstream during spring and summer. Seasonal meltwater sustains ecosystems, irrigates agriculture, and drives hydroelectric production. In Europe, the Alps are the most important water-supplying mountain range (Immerzeel et al., 2019). The Rhône, Danube, Rhine,

and Po basins are its main drainage basins and around 170 millions people reside around them (Mastrotheodoros et al., 2020).

To enable the assessment of climate change down to regional and local scales, the Global Climate Observation System program (Global Climate Observing System, 2006) set a requirement of at least 500 m horizontal resolution and 24 h temporal resolution to map snow area extent. Beyond climate monitoring, other applications require more detailed information about the snowpack, including snow depth and Snow Water Equivalent (SWE). These parameters are essential for managing hydroelectric power generation, irrigation systems, and water resources (Rouhier, 2018).

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These variables were originally estimated from ground measurements collected at meteorological stations or during field expeditions. While these methods provide highly accurate data, they are spatially limited. To assess snow dynamics at regional scales, geostatistical approaches have been used to interpolate between these sparse data points (Hosang and Dettwiler, 1991; Phillips et al., 1992; Erxleben et al., 2002).

Remote sensing methods overcome these limitations by providing regular observations over large spatial extents and have been extensively used for snow monitoring. Snow detection from optical data leverages the high reflectance of snow in the visible domain and its low reflectance in the Short-Wave InfraRed (SWIR) range (Dozier, 1984, 1989). Hall et al. (1995) introduced the Normalized Difference Snow Index (NDSI), which uses the reflectance contrast between these two spectral regions to identify snow-covered pixels based on a threshold value. Several studies developed strategies for NDSI-based segmentation by adding time and location-dependent thresholds (Poussin et al., 2023; Bousbaa et al., 2024), additional spectral screenings (Simon et al., 2018), or improved spectral indexes (Mohammadi et al., 2023). However, cloud cover often obstructs the surface. Even if the NDSI can separate clouds from snow, information about the ground is missing. To address this limitation, several approaches have been developed to fill cloud-induced gaps, including temporal interpolation (Hall et al., 2021), spatio-temporal interpolation (Li et al., 2020; Deng et al., 2024), optical sensor fusion (Moreno-Martínez et al., 2020; Sedano et al., 2014), and deep learning techniques (Tsardanidis et al., 2024; Wang et al., 2025).

Synthetic Aperture Radar (SAR), on the other hand, is an active remote sensing technique that illuminates the surface with electromagnetic waves and measures the backscattered signal. Using wavelengths from 0.1 to 100 cm allows it to penetrate clouds. SAR data has been investigated to map snow since the launch of ERS-1 in 1991. The backscattering of snow in SAR images depends mainly on its liquid water content. Depending on the available information some methods are limited to the detection of wet snow only, while advanced SAR modes or multimodality allows the detection of the snow cover. Throughout this paper, the term “snow” refers to both dry and wet snow unless explicitly stated otherwise.

Wet snow, which strongly attenuates backscattered SAR signal amplitude due to its liquid water content, can be mapped with a dual-polarization image by comparing the signal attenuation from wet snow with a snow-free reference image (Nagler and Rott, 2000; Nagler et al., 2016), and thresholding the attenuation value. Several approaches have been proposed to address the sensitivity of this threshold to local conditions. James et al. (2024) introduced dynamical thresholding techniques to improve classification. Machine learning algorithms have been explored: Gallet et al. (2024) compared several machine learning techniques to detect wet snow with labels from the snowpack model CROCUS (Brun et al., 1989). Lê et al. (2023) predict wet snow cover maps from Sentinel-1 backscatter amplitude and reference images with a Convolutional Neural Network (CNN), using labels derived from SAR images using Nagler’s method and coincident optical images. Gong et al. (2024) proposed a weakly supervised approach using Hierarchical Fuzzy C-means clustering on differential SAR images derived from Nagler’s method to generate pseudo-labels for training a dual-branch CNN for wet snow detection.

Despite dry snow being almost transparent to dual-polarization SAR signal, Lievens et al. (2019) proposed a method to retrieve dry snow depth with Sentinel-1 signal depolarization, though it requires prior information on snow presence which is obtained from Sentinel-2 optical data, although this method is limited to vegetation-free surfaces.

To detect the apparition of snow, encompassing wet and dry, Strozzini et al. (1999) and Singh et al. (2008) proposed InSAR-based change detection methods, that can be applied even in vegetated terrain. Quad-pol techniques were explored by He et al. (2017) to map snow cover. Tsai et al. (2019) mapped snow cover using dual-pol images

under the assumption of reciprocal targets (VH and HV are considered the same). Other approaches based on dual-polarization SAR images and deep learning techniques were developed to map snow cover. Montginoux et al. (2023) predicts snow cover maps from solely Sentinel-1 dual-pol backscatter using a CNN supervised with optical-derived labels. Daudt et al. (2023) uses both Sentinel-1 backscatter amplitude and Sentinel-2 optical data, along with elevation information, as input to a Recurrent Neural Network (RNN) to retrieve snow depth at weekly temporal resolution.

In this paper, we present a method to detect the snow cover from SAR dual-pol amplitude images only, using a CNN. To this end, we produce pseudo-labels from optical images, allowing us to detect jointly dry and wet snow without manual labeling. However, cloud cover limits optical labels availability for model training. We propose to use temporal interpolation to create pseudo-labels and increase the level of supervision. Unlike standard NDSI-based methods, our approach uses NDSI time series solely during training to supervise the model: at inference time, only SAR data is required, enabling snow cover estimation regardless of snow liquid water content and cloud conditions. To this aim,

- We present a dataset containing SLC coregistered SAR images over different orbits with snow cover labels obtained from optical images.
- We investigate whether a weakly supervised method relying on temporal interpolation and regularization to increase the amount of optical-derived labels and reduce label noise can improve the prediction of snow cover maps from SAR backscatter amplitude image.
- We show that both dry and wet snow can be detected jointly with similar performance from SAR dual-pol amplitude images using a deep-learning framework.

The CNN training framework is illustrated Fig. 1. In Section 2, we present the temporal interpolation methods and our weakly supervised deep learning framework. In Section 3, we present our study area and the data available, how we process SAR data and create our dataset. In Section 4, we present the model training and the different experiments. In Section 5 we present the results of each experiment and we discuss them in Section 6. Finally, we conclude in Section 7.

2. Methodology

Let $\mathbf{S} \in \mathbb{R}^{T_S \times H_S \times W_S \times 2}$ be the SAR time series and $\mathbf{O} \in [-1; 1]^{T_O \times H_O \times W_O}$ be the NDSI time series, over the same geographical area. The SAR series $\mathbf{S}$ contains co- and cross-polarized backscatter amplitudes (σ_{VV}^0 and σ_{VH}^0 respectively) in linear scale. Since NDSI and SAR data may differ in temporal and spatial resolutions, we denote T_O and T_S as the number of time steps in the NDSI and SAR time series, and H_O, W_O and H_S, W_S as their respective spatial dimensions (height and width). An index in the time series arrays $\mathbf{O}$ or $\mathbf{S}$ corresponds to a specific dimension (time, height, or width). Fixing one or more indices defines a position in the array and selects the corresponding value or subset. By fixing the temporal index t , we obtain a full spatial image at date t : $\mathbf{O}_t \in [-1; 1]^{H_O \times W_O}$ for NDSI data, and $\mathbf{S}_t \in \mathbb{R}^{H_S \times W_S \times 2}$ for SAR data. By fixing the spatial indices (h, w) , we obtain a temporal profile for that pixel: $\mathbf{o}_{h,w} \in [-1; 1]^{T_O}$ for NDSI, and $\mathbf{s}_{h,w} \in \mathbb{R}^{T_S \times 2}$ for SAR. Fixing all three indices (t, h, w) gives a single observation: $o_{t,h,w} \in [-1, 1]$ for NDSI, and $s_{t,h,w} \in \mathbb{R}^2$ for SAR.

Our method aims to train a Convolutional Neural Network (CNN) f to predict binary snow cover maps $\hat{\mathbf{Y}}_t \in \mathbb{B} = \{0, 1\}^{H_S \times W_S}$ from SAR imagery $\mathbf{S}_t \in \mathbb{R}^{H_S \times W_S \times 2}$. This poses a weak supervision problem: while we derive labels from the NDSI time series using a threshold (details in Section 2.1), many pixels $o_{h,w,t}$ are unobservable due to cloud cover. To address this, we propose a weakly-supervised framework where missing labels are estimated via temporal interpolation of each pixel time series $\mathbf{o}_{h,w} = (o_0, \dots, o_{T_O-1})_{h,w}$, generating pseudo-labels

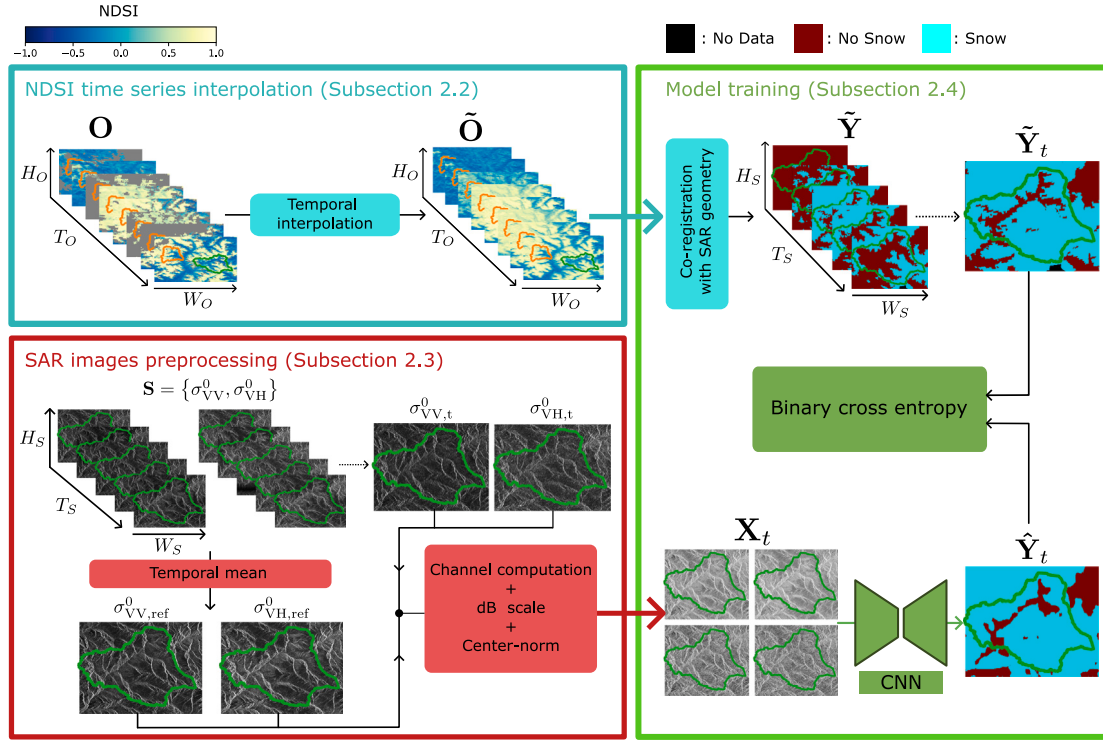


Fig. 1. Illustration of our training framework. Dotted arrows show the selection of one date among the time series. The optical NDSI time series $\mathbf{O}$ is temporally interpolated to obtain a cloud-gap filled time series $\tilde{\mathbf{O}}$. The interpolation methods are described in Section 2.2. The time series $\tilde{\mathbf{Y}}$ consists of binary snow cover pseudo-labels. It is obtained by thresholding the interpolated NDSI time series $\tilde{\mathbf{O}}$ and aligning it spatially and temporally onto the SAR time series $\mathbf{S}$. $\mathbf{S}$ contains backscatter amplitude values in VV and VH polarizations (σ_{VV}^0 and σ_{VH}^0 respectively). To create the SAR inputs, we first compute reference images for both polarizations $\sigma_{VV,ref}^0$ and $\sigma_{VH,ref}^0$. We estimate snow cover maps at time t by selecting σ_{VV}^0 and σ_{VH}^0 values from this date, $\sigma_{VV,t}^0$ and $\sigma_{VH,t}^0$, combine it to the reference images to build the input channels $\mathbf{X}_t$. The SAR data preprocessing and different channels combinations are described in Section 2.3. To train the model f , we minimize a binary cross entropy between the estimated snow cover map at time t $\tilde{\mathbf{Y}}_t$ and model prediction $\hat{\mathbf{Y}}_t = f(\mathbf{X}_t)$. The NDSI data is used only during training. The training framework is described in Section 2.4. For inference, only SAR data is needed to estimate snow cover maps.

for supervision. Two interpolation strategies are explored: a simple Closest Neighbours Interpolation (CNI) method, and a more advanced Kalman smoothing (KS), which estimates all missing observations and regularizes the estimated time series via a tunable noise variance ratio η . This second approach allows us to use the full SAR time series for training, despite the sparsity of the supervision signal.

Although both SAR and NDSI data are structured as time series, our CNN is applied independently for each date t , without modeling temporal dependencies. The SAR time series is used solely to build the multi-channel input $\mathbf{X}_t$. The NDSI time series serves only for supervision during training to generate labels, or pseudo-labels via interpolation.

In this section, we first introduce the NDSI and its uses. We then present the two temporal interpolation methods used to generate pseudo-labels for weak supervision, and finally we describe our model training framework.

2.1. Normalized difference snow index

Snow is highly reflective in the visible part of the electromagnetic spectrum and highly absorptive in the short-wave infrared domain, whereas cloud reflectance remains high in both spectral regions. Based on these spectral properties, the Normalized Difference Snow Index (NDSI) was introduced to distinguish snow from clouds in optical imagery by combining reflectance values from the visible (typically green) and SWIR bands (Dozier, 1984, Hall et al., 1995).

$$\text{NDSI} = \frac{\rho_{\text{Green}} - \rho_{\text{SWIR}}}{\rho_{\text{Green}} + \rho_{\text{SWIR}}} \quad (1)$$

where ρ denotes surface reflectance. NDSI ranges between -1 and 1 , the presence of snow in a pixel theoretically lead to a positive NDSI.

However (Hall et al., 1995) introduced a 0.4 threshold to assess if a pixel is snow covered or snow free. They empirically determined that if a pixel has a NDSI value above 0.4, more than half of its area is covered with snow. Several studies optimize the threshold which can be suboptimal for pixel in dark or forested areas or if it covers a too large area (Härer et al., 2018). Hall and Riggs (2021) uses additional data screens like altitude, surface temperature and other reflectance bands to filter out pixels that have a positive NDSI but no snow cover. The NDSI can be regressed to retrieve finer-grained information, such as fractional snow cover, the snow-covered fraction of a pixel area (Salomonson and Appel, 2004; Gascoin et al., 2020), Snow Water Equivalent or snow depth (Teleubay et al., 2023). In our method, we apply the traditional 0.4 threshold to derive binary snow cover labels.

2.2. Optical time series interpolation

2.2.1. Closest neighbours interpolation

As a simple temporal interpolation method, we perform linear interpolation to fill gaps in a three-days window, which is the mean duration of cloud obstruction (Hou et al., 2019).

$$\begin{cases} \tilde{o}_t = \frac{1}{2} (o_{t-1} + o_{t+1}) \\ \tilde{o}_t = o_{t-1} \text{ if } o_{t+1} \text{ missing} \\ \tilde{o}_t = o_{t+1} \text{ if } o_{t-1} \text{ missing} \\ \text{NaN else} \end{cases} \quad (2)$$

In the remaining of this paper, we will refer to this interpolation method as Closest Neighbours Interpolation (CNI). This method is close to the Cloud Gap Filled method used in the MOD10A1F dataset (Hall et al., 2021), which extends the last NDSI measurement until a new

observation is available. With CNI, future observations are also taken into account and we minimize the risk of missclassification by extending the measurement only to the next day. However, CNI will not fill all the gaps in the optical time series.

2.2.2. Kalman smoothing

The Kalman smoother (KS) is a recursive algorithm that estimates the successive states of a system given a linear state-space model (SSM) that describes the system dynamics and a series of observations, by minimizing a mean squared-error between predictions and observations (Kalman, 1960). Both model noise and observation noise are assumed to follow a normal distribution of mean zero and known variance. As the whole timeseries is available, both past and future observations are used to estimate the state of a pixel. A filtering phase is first applied to use past information, then a backward pass adds future observations information. Kalman filtering and smoothing considers that the state of the system is hidden and that information about the system is only available through noisy observations.

The relationship between hidden states and observation must be linear, as well as the hidden state dynamics between two timesteps. In our case, we obtain our binary labels from NDSI, so both the hidden state and the observations are NDSI. We denote the “true” hidden NDSI time series of a pixel at location (h, w) , $\mathbf{x}_{h,w} = (x_0, \dots, x_{T_0})$ and the associated series of observations $\mathbf{o}_{h,w} = (o_0, \dots, z_{T_0})$. Moreover, snow (and thus NDSI) dynamics are complex and depend on other variables such as wind speed, air temperature or slope. As we expect the NDSI value to be close between two consecutive dates, we model the NDSI dynamic as a random walk with a step sampled from a Gaussian distribution with known mean and variance v_{mod} , which controls how much the NDSI is allowed to evolve between two consecutive dates. The SSM is expressed as:

$$\begin{cases} x_{t+1} = x_t + \epsilon_t^{\text{mod}} \\ o_t = x_t + \epsilon_t^{\text{mes}} \end{cases} \quad (3)$$

ϵ_t^{mod} and ϵ_t^{mes} are the model and observation noise. Both are assumed to follow Gaussian distributions with zero mean and known variance, v_{mod} and v_{obs} respectively. We consider the model noise variance v_{mod} as constant, and use it to normalize the estimated variance. We define η the normalized observation noise variance and $\bar{v}$ the normalized estimated noise variance:

$$\eta = v_{\text{obs}}/v_{\text{mod}} \quad (4)$$

$$\bar{v} = v/v_{\text{mod}} \quad (5)$$

To interpolate the time series, the KS first predicts the next state using the SSM and all past information. It then corrects the prediction by integrating observations. Once the forward pass is done, a backward pass is performed to include future observations.

Prediction Step:

$$\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1} \quad (6)$$

$$\bar{v}_{t|t-1} = \bar{v}_{t-1|t-1} + 1 \quad (7)$$

$$\bar{s}_t = \bar{v}_{t|t-1} + \eta \quad (8)$$

Correction Step:

$$\bar{k}_t = \bar{v}_{t|t-1} \bar{s}_t^{-1} \quad (9)$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + \bar{k}_t v_{\text{mod}} (o_t - \hat{x}_{t|t-1}) \quad (10)$$

$$\bar{v}_{t|t} = \bar{v}_{t|t-1} - \bar{k}_t^2 \bar{s}_t \quad (11)$$

Smoothing Step:

$$c_t = \frac{\bar{v}_{t|t}}{\bar{v}_{t|t+1}} \quad (12)$$

$$\hat{x}_{t|T} = \hat{x}_{t|t} + c_t (\hat{x}_{t+1|T} - \hat{x}_{t|t}) \quad (13)$$

$$\bar{v}_{t|T} = \bar{v}_{t|t} + c_t^2 (\bar{v}_{t+1|T} - \bar{v}_{t|t}) \quad (14)$$

When observations are not available for the time step t , the correction step is skipped and the *a priori* estimate becomes the *a posteriori* estimate used for next step prediction. The estimated noise variance $\bar{v}_{t|t}$ keeps increasing the more timesteps without observation the time series has (Eq. (7)).

Only the ratio of the two variances $\eta = v_{\text{obs}}/v_{\text{mod}}$ determines the shape of the interpolated signal: it acts as a regularization parameter controlling the Kalman gain $\bar{k}_t$, which weights the respective contributions of model prediction and observations. The individual noise variances only affect the estimated state uncertainty $\bar{v}$, which we do not use in our method. Each pixel at position (h, w) at time t has an associated estimated mean $\hat{x}_{h,w,t}$ and variance $v_{h,w,t}$; we retain only the former and denote the resulting interpolated time series $\hat{\mathbf{O}} \in \mathbb{R}^{T_0 \times H_0 \times W_0}$. In our setting, we fix $v_{\text{mod}} = 0.01$ (1% of the maximum NDSI value) and tune $\eta \in [10^{-3}; 10^2]$, effectively tuning our confidence in the observations relative to the model. We initialize the smoother with $x_0 = o_0$, or $x_0 = 0$ if $o_0 = \text{NaN}$, and $v_0 = 0.01$.

2.3. SAR time series preprocessing

To produce the SAR inputs $\mathbf{X}_t$ used for snow cover prediction, we preprocess the SAR time series $\mathbf{S}$ to derive snow-informative features. All images are converted to dB scale and outlier values are saturated at the 0.5th and 99.5th percentiles to reduce speckle noise influence. We then compute reference images $\sigma_{\text{VV,ref}}^0$ and $\sigma_{\text{VH,ref}}^0$ from snow-free observations. This approach to improve SAR-based snow detection was introduced by Nagler and Rott (2000), and has become standard in SAR-based snow studies for detecting wet snow by comparing snow-covered acquisitions to snow-free reference scenes (Lê et al., 2023; James et al., 2024; Gallet et al., 2024; Karbou et al., 2021). Lievens et al. (2019) also used ratios with reference images for the estimation of dry snow height. Montginoux et al. (2023) builds on the dual-pol based methods of Nagler and Rott (2000) and Lievens et al. (2019), inputting their ratios in a CNN. Reference images are estimated from snow-free summer dates (from July to August) using a maximum likelihood estimator under a Rayleigh distribution assumption.

These reference images allow us to build channels which are less sensitive to local acquisition geometry. Different channel combinations such as raw backscatter amplitude, backscatter-to-reference ratios or concatenation, or snow-specific indices are compared in terms of their effectiveness for snow detection. The construction and selection of these SAR channels are detailed in Section 4.2.

2.4. Model training framework

We train a Convolutional Neural Network (CNN) f to predict binary snow cover maps $\hat{\mathbf{Y}}_t \in \mathbb{B}^{H_S \times W_S}$ for a given date t , using as input a single SAR image $\mathbf{X}_t \in \mathbb{R}^{H_S \times W_S \times C}$ derived from the SAR time series. Although the full SAR time series $\mathbf{S}$ is used during preprocessing to construct the input $\mathbf{X}_t$ to compute temporal reference values, the model operates independently on each date, the full time series is never input to the model.

$$\hat{\mathbf{Y}}_t = f(\mathbf{X}_t) \quad (15)$$

The model is supervised using pseudo-labels derived from the interpolated optical NDSI time series $\hat{\mathbf{O}}$. We first perform temporal alignment by retaining only the dates t where a SAR observation $\mathbf{S}_t$ is available. For each of these dates, we project the corresponding NDSI map $\hat{\mathbf{O}}_t$ onto the SAR geometry using georeferencing pixel information obtained by the projection of the RGE alti 1 m (Institut national de l'information géographique et forestière (IGN), 2021) in the French part of the image and the TINITALY 10m (Tarquini et al., 2007) in the Italian part of the image. When projecting the RGE alti 1 m, multiple DEM points are projected in the same pixel. The georeferencing information are obtained by averaging these points. On the Italian part of the image, the projection of the DEM leads to a sparse map that is

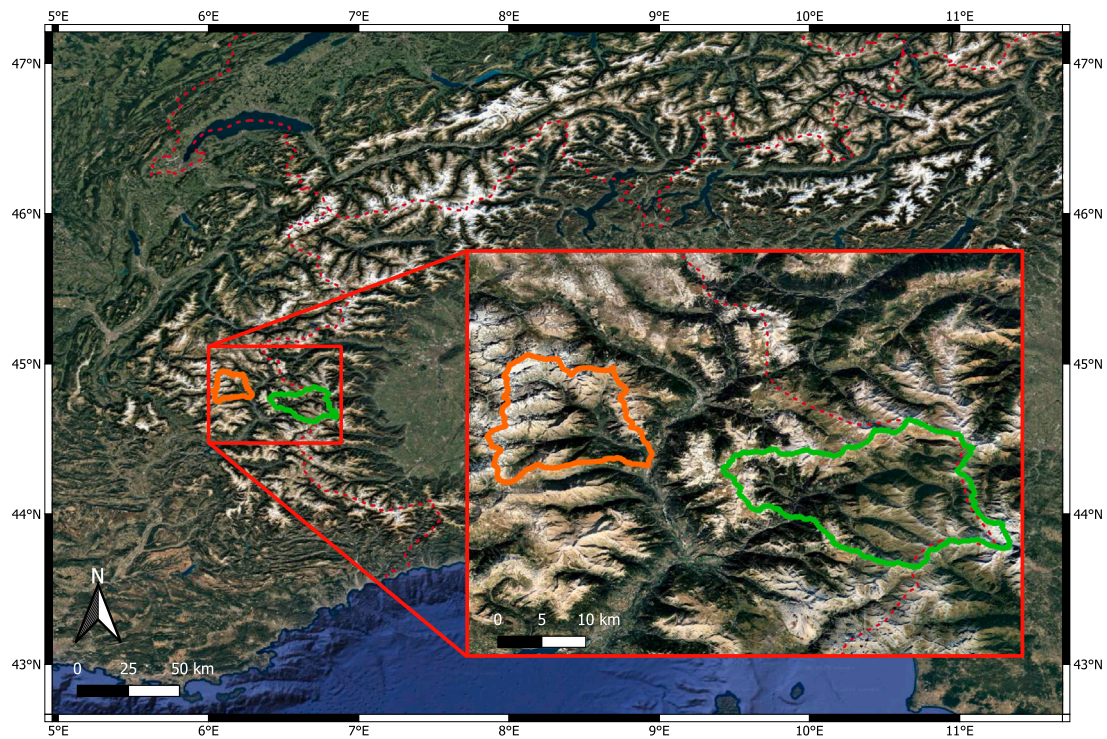


Fig. 2. Map of the two study basins in Rhône-Alpes region, France. —: Guil basin, —: Gyronde basin.

then interpolated. The aligned NDSI maps are then thresholded at 0.4 to generate a binary snow cover time series $\hat{\mathbf{Y}} \in \mathbb{B}^{T_S \times H_S \times W_S}$.

During training, the model receives a SAR input $\mathbf{X}_t$ for a given date t and predicts a binary snow cover map $\hat{\mathbf{Y}}_t$. The model is trained by minimizing the binary cross-entropy loss between prediction and target map $\hat{\mathbf{Y}}_t$:

$$\mathcal{L}_{BCE}(\hat{\mathbf{Y}}_t, \mathbf{Y}_t) = - \sum_{r=1}^{H_S \times W_S} [\tilde{y}_r \log(\hat{y}_r) + (1 - \tilde{y}_r) \log(1 - \hat{y}_r)] \quad (16)$$

We ignore pixels without labels when summing the loss. Note that the optical data is used only during training to provide supervision. Once trained, the model predicts snow cover maps only from SAR data. The hyperparameters we consider are: the learning rate, the number of training epochs, the classification threshold and the value of the regularization parameter η when using KS for label interpolation. We describe their optimization in Section 4.1.

3. Data and study area

3.1. Study area

We study the Guil and Gyronde river basins located in the French Alps near the Italian border (illustrated in Fig. 2) during the 2018 and 2019 study years, defined here as running from July 1st, 2018 to June 30th, 2019, and from July 1st, 2019 to June 30th, 2020, respectively. Each period includes the summer preceding the water year (which spans from September 1st to August 31st), allowing us to capture pre-season conditions relevant for snow and soil dynamics. The Guil basin covers an area of 411 km², while the Gyronde basin covers an area of 230 km². Fig. 3 illustrates the elevation profiles of both basins. The Guil basin has an altitude ranging homogeneously from 1000 m to 4000 m while Gyronde basin exhibits steeper slope. In both basins, elevations are predominantly concentrated around 2500 m. This mid-altitude configuration leads in complex snow dynamics. At these elevations, temperatures frequently fluctuate around the freezing point, leading to multiple freeze-thaw cycles. These cycles influence the

snowpack stability, soil moisture and root growth (Man et al., 2022). Consequently, our remote sensing method needs to be able to map both wet and dry snow to allow precise alpine environmental dynamics studies.

In this study, we use data from the Guil basin for both 2018 and 2019 periods, and from the Gyronde basin for 2018 only. The Guil 2018 time series is used as training and validation sets, and test set without domain shift. The Guil 2019 and Gyronde 2018 are used as two other test sets to test our method robustness to temporal and spatial domain shift.

3.2. Sentinel-1 SAR images

We use Sentinel-1 SLC IW (Single Look Complex - Interferometric Wide swath) data acquired by the Sentinel-1 A and Sentinel-1B satellites within the European Space Agency's Copernicus program. The acquisitions we use are performed in C-band (~5.4 GHz), and provide dual-polarization (VV and VH) acquisitions with a spatial resolution of approximately 5 × 20 m in range-azimuth. The temporal revisit frequency of 6 days allows for regular monitoring of snow cover dynamics. The images are deburst and coregistered with an interferometric precision using LabSAR algorithm (Weissgerber et al., 2022). Three Sentinel-1 orbits cover the Guil basin: two descending orbits D139 and D66, and one ascending orbit A88. The Gyronde basin is also covered by three orbits: descending orbit D139 and ascending orbits A88 and A161. Therefore, we have 3 acquisitions every 6 days for each basin. The spatial dimensions (H_S, W_S) of the Sentinel-1 images vary across acquisition dates, as they depend on the specific orbit and imaging geometry. As a result, each time series is composed of temporally ordered SAR images with date-dependent spatial extents. Table 1 summarizes the Sentinel-1 acquisitions used for each time series.

3.3. Snow products

3.3.1. MODIS NDSI

The MODIS instrument provides multispectral observations every 1 to 2 days over the globe. We use the MOD10A1 dataset (Hall and

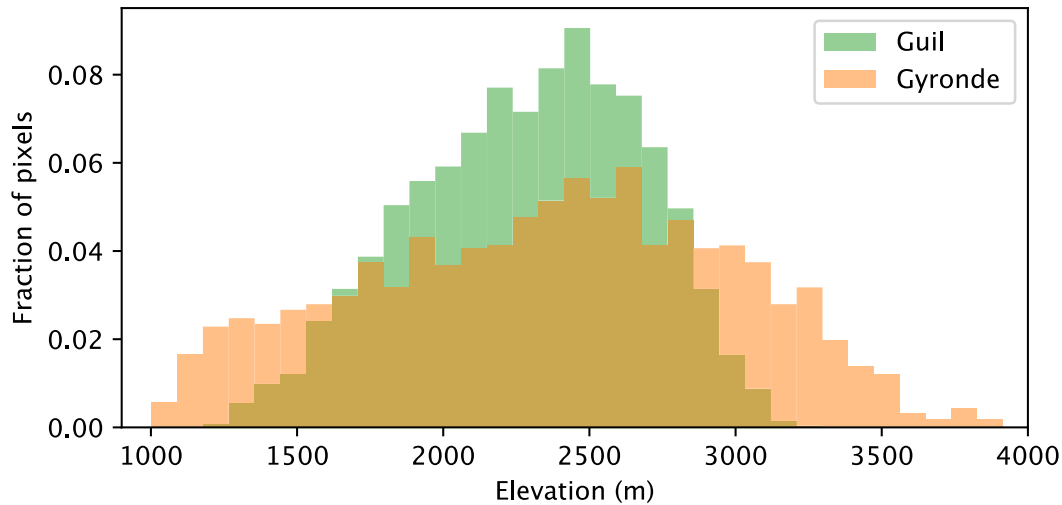


Fig. 3. Elevation profile of Guil and Gyronde basins. ■: Guil, ■: Gyronde,

Table 1

Sentinel-1 acquisitions used over the Guil and Gyronde basins during the 2018 and 2019 study periods. D stands for “Descending” and A for “Ascending”.

Basin and year	Orbit	Number of acquisitions	Spatial dimensions (Height × Width in pixels)
Guil 2018	D139	55	1872 × 8889
	D66	55	1996 × 7833
	A88	54	1934 × 10745
Guil 2019	D139	56	1872 × 8889
	D66	56	1996 × 7833
	A88	54	1934 × 10745
Gyronde 2018	D139	54	2038 × 4214
	A88	54	1964 × 5107
	A161	28	2147 × 6019

Riggs, 2021), which provides daily NDSI measurements over our study area at 500×500 m resolution at nadir. The one day revisit time ensures that the labels are always acquired the same day as each SAR acquisition, enhancing their reliability in the face of the fast dynamics of snow cover. The NDSI product is a composite of several acquisitions of the day. In addition to NDSI, the dataset provides a cloud mask. The binary labels are obtained by thresholding the NDSI at 0.4. For the pseudo-labels, the thresholding is applied after the NDSI time series interpolation.

3.3.2. THEIA L2B snow product

The THEIA L2B Snow product provides a binary snow cover product and a cloud mask based on Sentinel-2 and Landsat-8 data at 20 m resolution. An acquisition is available every 5 or 10 days. The snow product is based on the Let-it-snow (Simon et al., 2018) processing chain, which takes as input Sentinel-2 and Landsat-8 surface reflectances corrected for atmospheric and slope effects, a digital elevation model and the associated cloud mask from THEIA L2 A product. The Sentinel-2 tiles covering our study area are 32TLQ and 31TGK.

THEIA L2B Snow product is only used to validate our method, and to assess the transfer of the snow detection capacity from optical images to SAR images.

3.3.3. Copernicus wet/dry snow product

The Copernicus Wet/Dry Snow (WDS) product provides snow state classification over Europe in the entire EE38+UK area at 60 m resolution. This dataset is based on overlapping acquisitions from Sentinel-1 and Sentinel-2 at the same date. To detect snow, it uses the Sentinel-2 Fractional Snow Cover on the Top Of Canopy (FSCTOC). Binary snow cover extent is retrieved from FSCTOC by applying a threshold

at 60%. To differentiate wet snow from dry snow, it uses the SAR-based algorithm on Sentinel-1 data to map snowmelt area extent with Nagler’s method (Nagler et al., 2016). A pixel is classified as wet snow when detected with S1 wet snow algorithm and $FSCTOC \geq 60\%$. When $FSCTOC \geq 60\%$ and no wet snow is detected with S1, the pixel is classified as dry snow. Forested areas, urban areas and water areas are masked using the Copernicus High Resolution Layers Tree Cover Density (TCD) of the year 2018, Imperviousness Density (IMD) of the year 2018 and the Water Layer of 2018 because the wet snow detection was deemed unreliable. Clouds, clouds shadows and regions affected by SAR shadows, foreshortening and layover are also masked.

Copernicus WDS products is not used for training our model, but to validate our model ability to detect both wet and dry snow.

4. Experimental settings

4.1. Model training and evaluation

We train the network using Guil basin acquisitions between September 1st, 2018 and June 1st, 2019. To compute reference images, we use acquisitions from July 1st to August 31st, 2018, as detailed in Section 4.2. Out of the 134 acquisitions available across the three orbits, we use 88 dates for training, 24 for validation, and 22 for testing. Fig. 4 shows the split into train, validation, and test sets over the 2018 water year.

We chose the test dates so that a Sentinel-2 acquisition, and thus a label map from THEIA L2B Snow or Copernicus WDS, is also available. Since THEIA L2B Snow and Copernicus WDS products are available every 5 to 10 days, we regularly sample the water year ensuring that our test set is representative of the whole water year.

Table 2

Training hyperparameters per configuration. A configuration is defined by a combination of interpolation method and channel set. Channel sets are described in Table 3. ES: early stopping. The classification threshold and, for KS models, the regularization parameter η , are selected per model and per configuration respectively by maximizing the F1-score on the validation set.

Interpolation	Channel set	Optimizer	Learning rate	Batch size	Crop size	ES patience	ES δ
None	A, B	Adam	5×10^{-5}	8	512×512	5 epochs	0.01
	C, D		10^{-5}				
KS	B		10^{-3}				
CNI	B		10^{-3}				

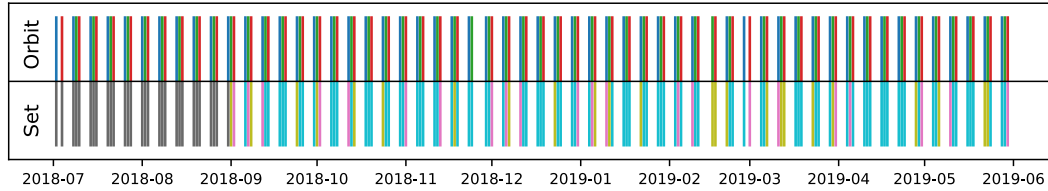


Fig. 4. Guil 2018 time series split into train, validation, and test sets. Orbits are shown with bar color above the middle line and sets with bar colors below. Test dates also have a Sentinel-2 observation available. Orbits: D139, A88, D66. Sets: reference dates, train dates, validation dates, test dates.

We extract 512×512 patches from the 88 training images using a regular grid with an overlap of 256 pixels to increase the amount of training data. A U-Net architecture (Ronneberger et al., 2015) with an EfficientNet-B0 encoder is trained using the fixed hyperparameters listed in Table 2. Interpolated labels are used for training only. For validation, non-overlapping 512×512 patches are extracted from the 24 validation images. We did not use image augmentations such as rotations or histogram transformations to preserve the statistics and physics of SAR images.

Each model is characterized by the set of SAR channels used as inputs and the interpolation method used to label the SAR pixels. For each configuration, we train 6 models. After training, the classification threshold is selected independently for each model by maximizing the overall F1-score on the validation set. For models trained with KS interpolated labels, the regularization parameter η is also an hyperparameter that is similarly selected per configuration by maximizing the Overall-F1 on the validation set.

Predictions are performed over the entire image domain $H_S \times W_S$ using overlapping 512×512 patches with a stride of 128 pixels. Each pixel is predicted multiple times with varying context, and final predictions are obtained by Gaussian-weighted averaging, giving more importance to central pixels.

4.2. Selection of SAR channels

Sentinel-1 SLC images are preprocessed before being used as input to the CNN. We discard the phase and retain only the amplitude-related data. All preprocessing steps described in Section 2.3 are applied independently for each polarization, orbit, basin and water year. To compute these reference images, we select all the dates between July 1st and August 31st and estimate the reference images $\sigma_{VV,ref}^0$ and $\sigma_{VH,ref}^0$ with the maximum likelihood estimator under a Rayleigh distribution assumption.

We compare four sets of SAR channels described in Table 3. The set A concatenates the two base polarimetric channels. The set B adds reference images. The set C, also used in L   et al. (2023) and Gallet et al. (2024) concatenates ratios of backscatter channels with their references.

$$\sigma_{VV,ratio}^0 = \sigma_{VV}^0 / \sigma_{VV,ref}^0 \quad (17)$$

$$\sigma_{VH,ratio}^0 = \sigma_{VH}^0 / \sigma_{VH,ref}^0 \quad (18)$$

Using ratios allows to be independent from local incident angle as we measure a difference from the snow-free response for each pixel.

Table 3

Description of the compared set of SAR channels.

ID	Concatenated channels
A	$\{\sigma_{VV}^0, \sigma_{VH}^0\}$
B	$\{\sigma_{VV}^0, \sigma_{VH}^0, \sigma_{VV,ref}^0, \sigma_{VH,ref}^0\}$
C	$\{\sigma_{VV,ratio}^0, \sigma_{VH,ratio}^0\}$
D	$\{R_{dry}, R_{wet}\}$

The set D used in Montginoux et al. (2023) concatenates specialized channels for wet and dry snow detection R_{wet} and R_{dry} .

$$R_{wet} = \frac{1}{2} \sigma_{VV,ratio}^0 + \frac{1}{2} \sigma_{VH,ratio}^0 \quad (19)$$

$$R_{dry} = \frac{\sigma_{VH}^0 / \sigma_{VV}^0}{\sigma_{VV,ref}^0 / \sigma_{VH,ref}^0} \quad (20)$$

The channel R_{wet} comes from Nagler's method (Nagler and Rott, 2000), with a constant factor 0.5 to weight VV and VH polarization equally. The channel R_{dry} is an adaptation of the one introduced in Lievens et al. (2019). The ratio between VV and VH polarizations allows to assess the depolarization caused by the dry snow pack. These channels are optimized for wet and dry snow detection, we want to assess if the network needs specialized channels or finds better features itself for snow detection. Table 3 summarizes the channels sets.

For each input channel, we compute the mean and standard deviation over the entire time series to standardize the data.

4.3. Robustness to domain shift: spatial and temporal transfer

Deep learning models performance can decrease when generalizing outside of their training domain. However, training data is limited. Our goal is to produce a model that does not need to be retrained for each study area and can generalize temporally. The training is done only using the SAR and NDSI time series of the 2018 water year over the Guil basin. To evaluate if our method produces models robust to temporal or spatial transfer we use additional test sets. To test the robustness to temporal transfer, we include the SAR and NDSI time series of the 2019 water year over the Guil basin. As to test the robustness to spatial transfer we include the SAR and NDSI time series of the 2018 water year over the Gyrone basin.

The evaluation on MODIS NDSI, THEIA L2B Snow and Copernicus WDS labels will be performed on each three domains.

4.4. Evaluation metrics

4.4.1. Joint detection of wet and dry snow

For quantitative model evaluation on MODIS NDSI and THEIA L2B snow products, we compute the cumulated confusion matrix on all dates of the test time series (*Snow* is the positive class). Accuracy measures the proportion of correct predictions for both classes. Precision is the amount of correct predictions among all predictions of a class, recall is the amount of correct predictions among all actual instances of a class. The F1-score is the harmonic mean between precision and recall, and the Overall F1-score is the average of F1-score for positive and negative class weighted by the frequency of each class. Because we use the cumulated confusion matrix on the whole time series our F1-score is close to a micro-averaged F1-score.

$$\begin{aligned} \text{Precision}_{\text{Snow}} &= \frac{TP}{TP + FP} \\ \text{Recall}_{\text{Snow}} &= \frac{TP}{TP + FN} \\ \text{Precision}_{\text{No Snow}} &= \frac{TN}{TN + FP} \\ \text{Recall}_{\text{No Snow}} &= \frac{TN}{TN + FN} \end{aligned} \quad (21)$$

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + FP + TN + FN} \\ \text{F1-score} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{Overall F1} &= f_{\text{Snow}} \times \text{F1-score}_{\text{Snow}} + \\ &\quad (1 - f_{\text{Snow}}) \times \text{F1-score}_{\text{No Snow}} \end{aligned} \quad (22)$$

with TP the number of true positives, FP the false positives, TN the true negatives and FN the false negatives. Model scores are given as mean and standard deviation, computed among the 6 models with identical configurations.

4.4.2. Wet and dry snow

As our model predicts binary snow, we cannot compute all the metrics on wet and dry snow when evaluating on Copernicus WDS product. We consider a pixel as true positive when its label is wet or dry snow and the model predicts it as snow covered. We consider a pixel as false negative when its label is wet and dry snow and the model predicts it as snow free.

$$\begin{aligned} \text{TP}_{\text{wet}} &= \sum_i [(\hat{y}_i = \text{Snow}) \wedge (y_i = \text{WetSnow})] \\ \text{FN}_{\text{wet}} &= \sum_i [(\hat{y}_i \neq \text{Snow}) \wedge (y_i = \text{WetSnow})] \\ \text{TP}_{\text{dry}} &= \sum_i [(\hat{y}_i = \text{Snow}) \wedge (y_i = \text{DrySnow})] \\ \text{FN}_{\text{dry}} &= \sum_i [(\hat{y}_i \neq \text{Snow}) \wedge (y_i = \text{DrySnow})] \end{aligned} \quad (23)$$

We cannot compute false positives and true negatives for dry and wet snow classes. We can thus only evaluate our model recall on the following classes:

$$\begin{aligned} \text{Recall}_{\text{wet}} &= \frac{\text{TP}_{\text{wet}}}{\text{TP}_{\text{wet}} + \text{FN}_{\text{wet}}} \\ \text{Recall}_{\text{dry}} &= \frac{\text{TP}_{\text{dry}}}{\text{TP}_{\text{dry}} + \text{FN}_{\text{dry}}} \end{aligned} \quad (24)$$

5. Results

5.1. Selection of SAR channels

Table 4 shows model performance on Guil 2018 test set. The mean and standard deviation are computed by evaluating metrics on 6 models trained with different seeds. This variability helps assess the robustness of the models and avoids overestimating performance due to a favorable seed. Using only the two backscatter amplitude channels

σ_{VV}^0 and σ_{VH}^0 (channel set **A**) yields an accuracy above 0.87, an F1-score of 0.897 for the class *Snow* and 0.831 for the class *NoSnow*. These results demonstrate that our method can accurately predict both dry and wet snow cover maps only from SAR data. Adding reference from means of snow-free images significantly improves accuracies and F1 for all models, whether it is concatenated with the raw backscatter amplitude σ_{VV}^0 and σ_{VH}^0 (channel set **B**) or combined with a ratio (**C** and **D**). Channel set **B** improves performance by 3 points on all metrics. Channel sets **C** and **D** yield smaller scores, but still improve from **A**. Sets **A**, **B**, and **C** show similar variance, while set **D** introduces slightly higher variability.

For qualitative evaluation, we look at model predictions on the Guil basin on March 31st 2019 in Fig. 5. All channel sets allow models to distinguish the majority of snow-covered from snow-free regions. Channel set **A** contains some speckle noise and the predicted snow cover map misses some *NoSnow* patches. Channel sets **C** and **D** are highly noised, and lead to poor segmentation with snow cover maps showing scattered and noisy predictions, especially in the center of the basin. Whereas channel set **A** yields compact patches, predictions from sets **C** and **D** appear more granular and result in numerous localized misclassifications. The ground truth shows that the spatial distribution of snow cover is correlated to the terrain elevation as intuition would suggest. **C** and **D** use ratios between backscatter amplitude channels and their references to create slope-independent features, but remove topographical information. While this can help reduce terrain-related variability, it tends to remove useful topographic information and leads to imprecise segmentation. We see that Channel set **B**, which includes reference images without ratios, allows the best segmentation, even if the *NoSnow* class is slightly over-predicted.

Based on both quantitative metrics and qualitative analysis, we select SAR channel set **B** for the remaining experiments.

5.2. NDSI interpolation

Before analyzing the impact of interpolation on model performance, we first consider the NDSI signal in the MODIS geometry (*i.e. before projection on SAR geometry*).

Fig. 6 compares the interpolated NDSI time series of a MODIS pixel obtained using the CNI method and the Kalman Smoother (KS) for a range of η values, along with the corresponding estimated variance. Blue dots represent the NDSI values estimated by the CNI method. As expected from its design, CNI keeps the interpolated values close to the original observations by filling only short gaps (up to 3 days) using linear interpolation or by extending the nearest valid measurement. The lines show the NDSI interpolated by the KS for different η values. In addition to gap-filling, the Kalman smoother changes the observed NDSI values, which can shift the class distribution in the training set. Staying too close to this distribution may indicate that observation noise was fully retained, while diverging too far suggests excessive regularization, removing relevant dynamics from the signal. For $\eta = 0.1$ (yellow), the interpolated time series is close to what would be obtained using simple linear interpolation between observations. This is consistent with the linear nature of the KS, where a small η implies high confidence in the observations. As η increases ($\eta \geq 10$), the signal becomes more heavily regularized: sharp increases in the NDSI signal, such as those observed in late November or April, are significantly smoothed. In mid-January, fluctuations around the classification threshold cause multiple class changes in the observations, particularly under strong regularization ($\eta = 100$, brown).

Estimation variance grows with longer gaps between observations: the range between minimum and maximum variance narrows, while the average variance increases. This behavior is consistent with high $\eta = \frac{\nu_{\text{obs}}}{\nu_{\text{mod}}}$ values, which reflects greater observation noise variance and results in a smaller Kalman gain k_t . Consequently, the observations are given less weight, and the state estimate relies more on the random walk model (Eq. (3)).

Table 4

Mean performance and associated standard deviations of models trained with different SAR channel sets on the Guil 2018 test set.

SAR channel set	Accuracy $\uparrow$	Overall F1 $\uparrow$	F1 _{Snow} $\uparrow$	F1 _{NoSnow} $\uparrow$
A: $\{\sigma_{VV}^0, \sigma_{VH}^0\}$	0.872 (0.0043)	0.873 (0.0042)	0.897 (0.0038)	0.831 (0.0054)
B: $\{\sigma_{VV}^0, \sigma_{VH}^0, \sigma_{VV,ref}^0, \sigma_{VH,ref}^0\}$	0.901 (0.0044)	0.902 (0.0042)	0.921 (0.0041)	0.867 (0.0044)
C: $\{\sigma_{VV,ref}^0, \sigma_{VH,ref}^0\}$	0.876 (0.0042)	0.876 (0.0039)	0.901 (0.0038)	0.832 (0.0044)
D: $\{R_{dry}, R_{wet}\}$	0.876 (0.0080)	0.877 (0.0077)	0.901 (0.0070)	0.835 (0.0090)

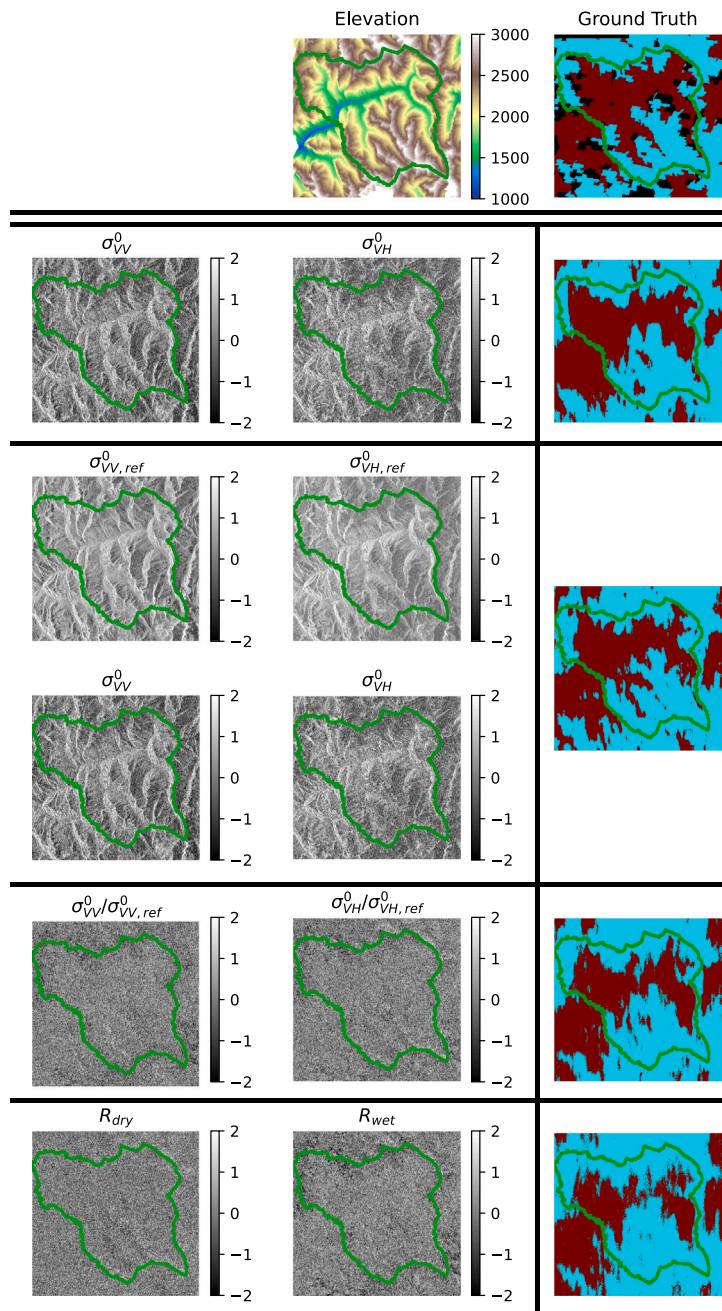


Fig. 5. Predictions in the Guil basin from acquisitions on March 31st, 2019, from models trained with different SAR channels. The ground truth is displayed in the top-right corner. SAR channels are centered and normalized in dB, predicted snow maps are in the right column. —: Guil basin, ■: Snow, ■: No Snow, ■: NoData.

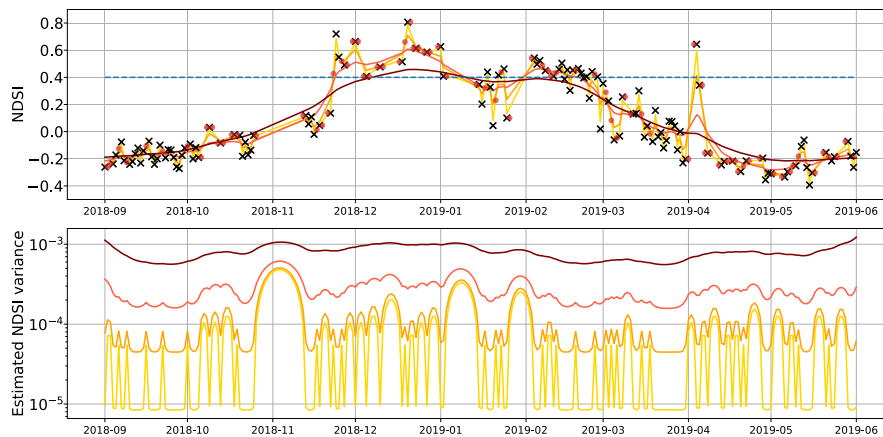


Fig. 6. Interpolated NDSI time series of a pixel in the Guil basin from 2018 September 1st to 2019 June 1st, using CNI and KS with different η values. Smoothed time series for different η values are shown in different colors. The upper graph shows the estimated values, the lower graph shows the KS estimation variance. The pixel elevation is 2026 m. —: NDSI threshold, x: Observations, •: CNI predictions, —: KS $_{\eta=0.1}$, —: KS $_{\eta=1}$, —: KS $_{\eta=10}$, —: KS $_{\eta=100}$.

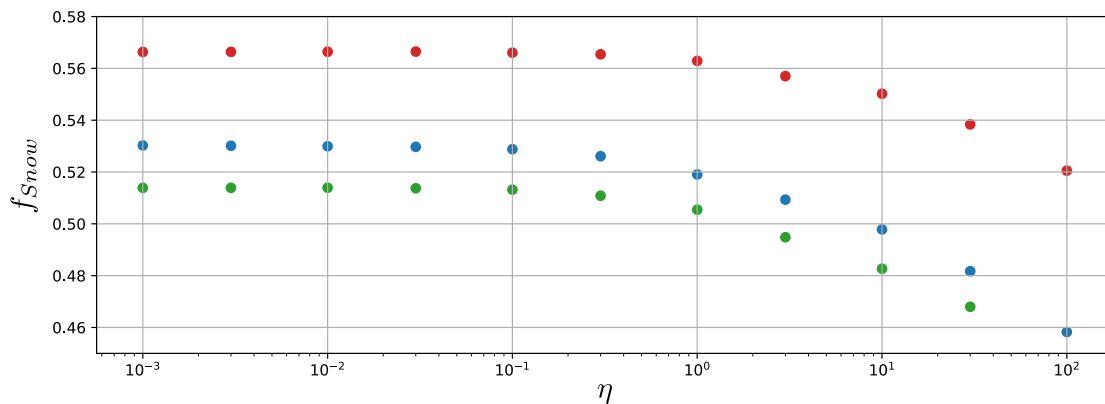


Fig. 7. Snow fraction in the Guil 2018 total set (train, validation and test) for each orbit as a function of the regularization parameter η . •: Descending 139, •: Ascending 88, •: Descending 66.

5.3. Impact of the interpolation in the training labels

In this section, we assess how thresholding and regularization affects the training labels, after projection onto the SAR geometry. To select the optimal level of regularization for KS without accessing a ground truth, we treat η as a hyperparameter and train models with various η values. Their performance is then evaluated on a non-interpolated validation set, which is reserved for this purpose.

5.3.1. Changes in class distribution

We show in Table 5 the impact of the two interpolation methods on labels distribution in the training set. The CNI method is enough to reduce by half the amount of missing labels. However, it also introduces a shift in the class distribution from observations by increasing the occurrence of the *Snow* class. This shift is likely due to the fact that snowfall events occur with cloud cover. As fresh snow exhibits a high NDSI, due to its low liquid water content and strong reflectance in the visible spectrum, the interpolation tends to yield elevated NDSI values, favoring *Snow* classification. KS also increases the number of *Snow* pixels, but stronger regularization dampens this effect. We show in Fig. 7 the shift in class distribution introduced by the KS regularization for a range of η values. $\eta \leq 10^{-3}$ does not introduce decision changes from raw observations. For $\eta \geq 1$, the snow fraction in the training labels changes more than 1%. We show in Fig. 8 how these classification decisions change the snow cover maps for a range of η values. We see that most changes occur in the boundary between snow covered and snow-free areas, suggesting that the Kalman smoother primarily affects uncertain pixels near the snow line.

5.3.2. Impact of the interpolation methods on models performance

Fig. 9 shows the accuracy and F1-scores of models trained with labels interpolated using the KS, for $\eta \in [10^{-3}; 10^2]$. Performance increases with η , reaching its highest value at $\eta = 0.3$. Beyond this point, mean performance trend is unclear and the variance increases. Since mean performance is maximum for $\eta = 0.3$, we select this value as the optimal η for the rest of the experiments. We refer to models trained without any interpolated labels as $\mathcal{M}_{\text{None}}$, which serve as our baseline. Models trained with CNI-interpolated labels are denoted $\mathcal{M}_{\text{CNI}}$, while $\mathcal{M}_{\text{KS}}$ refers to models trained with KS-interpolated labels.

Table 6 shows the evaluation on MODIS NDSI labels on all three test sets. The models are trained on pseudo-labels from each of our interpolation methods: our baseline with no interpolation, CNI and KS with $\eta = 0.3$. On the Guil 2018 test set, label interpolation increases both accuracy and Overall F1-score by approximately 2%, whether CNI or KS is used. $\mathcal{M}_{\text{CNI}}$ has a higher variance, likely due to tendency of CNI to introduce label noise by extending the last valid label in a pixel time series. $\mathcal{M}_{\text{KS}}$ yields slightly better average performance than $\mathcal{M}_{\text{CNI}}$, although the gain from no interpolation is smaller in comparison (less than 1% gain).

For temporal transfer, label interpolation also improves model performance. Moreover, all models, including our baseline, achieve higher scores than when tested on the same domain as training, we address this in Section 6.2.6. KS interpolation on labels still yields better results than CNI. In the Guil 2019 set, every date of the time series is used to evaluate the model instead of a subsample of the 2018 water year, which can explain the increase of performance from the test without transfer.

Table 5

Class distribution on train, validation and tests sets after projection on SAR geometry for different interpolations.

Set	Interpolation	NoData (%)	Snow (%)	No Snow (%)	f_{Snow}/f_{NoSnow}
Train	None	61.3	19.1	19.6	0.97
	CNI	27.7	39.2	33.1	1.18
	KS $_{\eta=1}$	2.4	51.5	46.1	1.11
	KS $_{\eta=10}$	2.4	49.7	47.9	1.03
Validation	None	27.3	37.7	35.0	1.07
Test	None				
Guil 2018		36.0	37.7	26.3	1.43
Guil 2019		49.28	29.02	21.70	1.34
Gyronde 2018		46.09	33.23	20.68	1.61

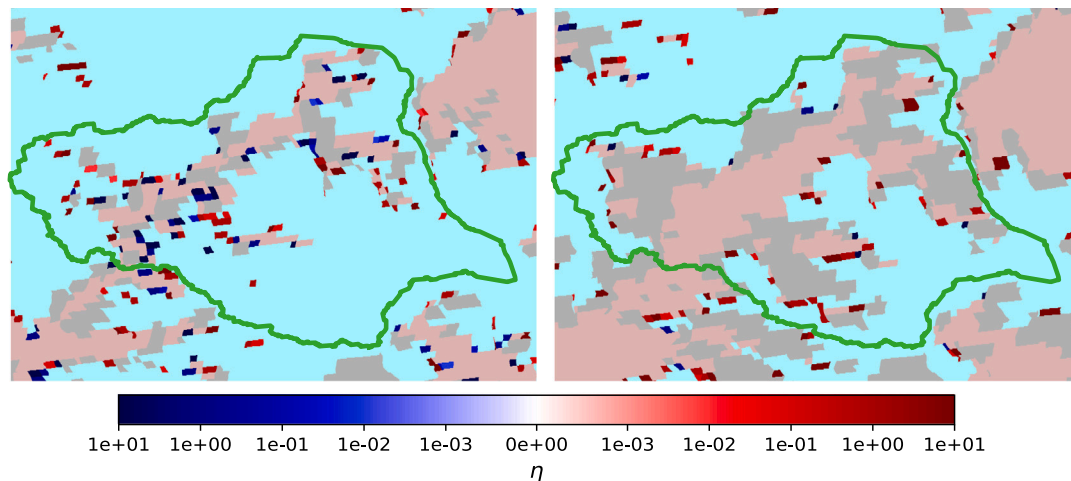


Fig. 8. Sensitivity of interpolated labels to the Kalman smoother regularization parameter η , illustrated for two SAR acquisition dates: February 22nd 2019 (left) and March 12th 2019 (right). The background shows the uninterpolated binary labels (light blue: Snow, light red: No Snow, grey: NoData). Colored pixels indicate label changes induced by the smoother: red pixels are Snow pixels reclassified as No Snow, blue pixels are No Snow pixels reclassified as Snow. Color intensity encodes the smallest η triggering the change (log scale). The green line shows the Guil basin.

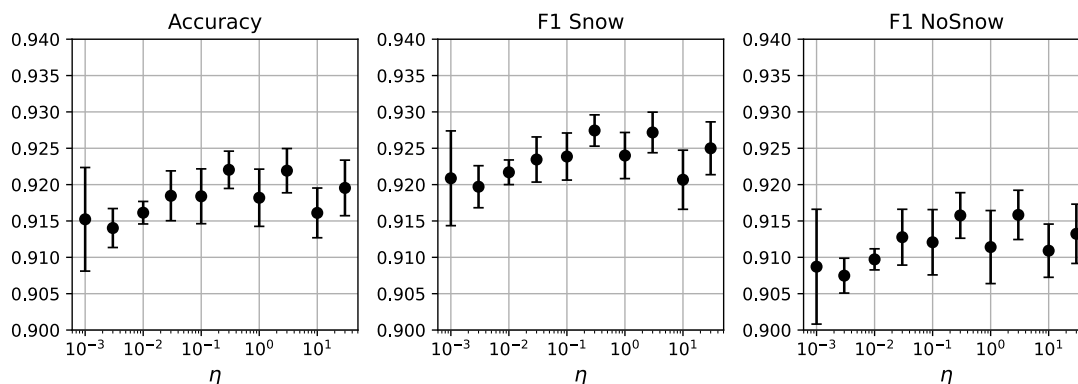


Fig. 9. Performance of models trained with KS interpolated labels with different η values on validation set. Mean scores are shown with a dot and standard deviation with a line.

For spatial transfer, mean performance decrease across all models, with increased associated variance. As for the two previous domains, increasing the amount of pseudo-supervision through label interpolation improves predictions. $\mathcal{M}_{CNI}$ and $\mathcal{M}_{KS}$ yield similar average performance, with $\mathcal{M}_{KS}$ exhibiting slightly lower variance. The interpolation method does not impact model performance, but still improves from the baseline.

Fig. 10 shows model predictions on the same domain, with temporal transfer and spatial transfer, respectively. Across all domains, label interpolation leads to smoother predicted snow cover maps and improves recall on the Snow class. In the case without transfer shown

in Fig. 10, $\mathcal{M}_{CNI}$ slightly outperforms $\mathcal{M}_{KS}$, but the resulting maps are visually similar and closely match the observations. Model predictions on temporal transfer scenarios also lead to snow cover maps close to observations. In the case of spatial transfer however, the predicted snow cover maps are less satisfactory. Models all predict the South-East area of the basin as NoSnow, and similar errors persist across multiple dates (not shown in the article). The drop in performance comes from entire patches that are missclassified instead of imprecise segmentation at the interface between classes, unlike in the Guil basin. The main snow-covered areas, however, are still correctly identified.

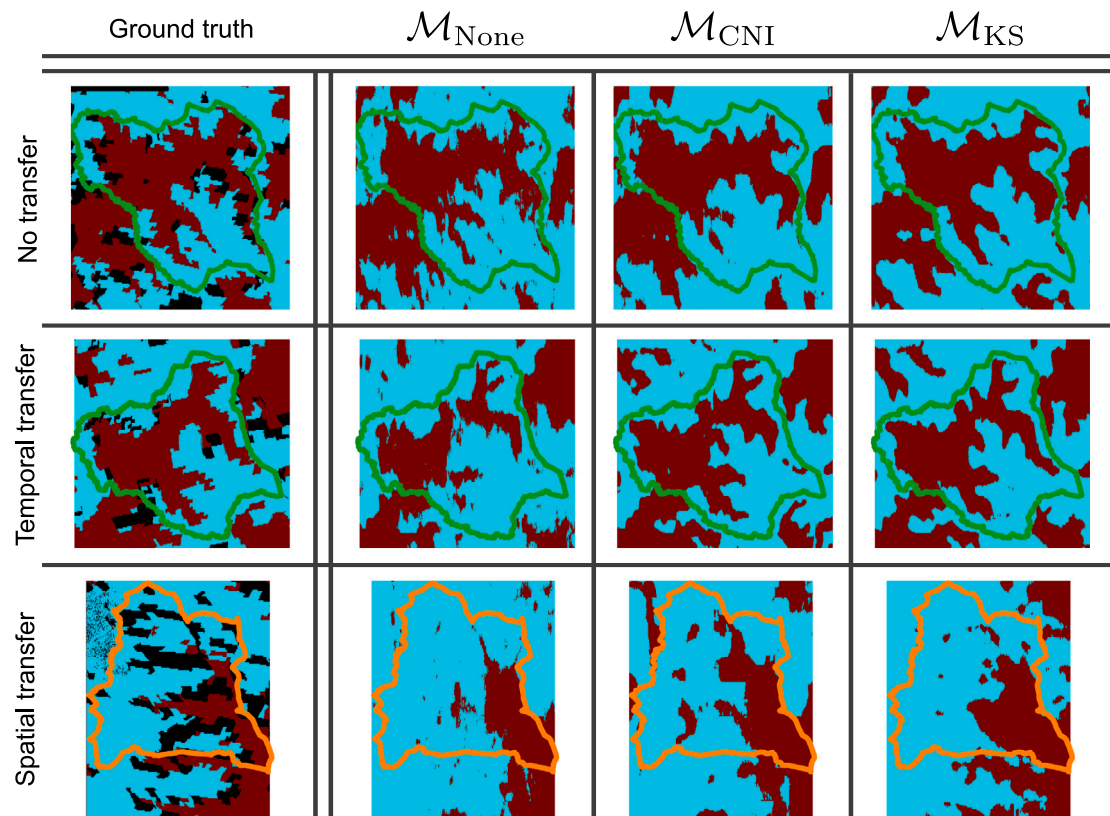


Fig. 10. Prediction of models trained with labels from different interpolation method, without domain shift, and under temporal and spatial shift. First row shows Guil basin on 2019 March 31st, second row shows Guil basin on 2020 April 10th and third row Gyrone basin on 2019 March 19th. —: Guil basin, —: Gyrone basin, ■: Snow, ■: No Snow, ■: NoData.

Table 6

Mean performance and standard deviations of models trained with pseudo-labels from different interpolation methods on three test sets, using MODIS NDSI labels. The first evaluates without domain shift, the second with temporal domain shift, and the third with spatial domain shift. Best mean scores are shown in bold; when two scores are tied, the one with the lower variance is underlined.

Interpolation	Accuracy ↑	Overall F1 ↑	F1-Snow ↑	F1-No Snow ↑
Test Guil 2018 (Same domain)				
None	0.901 (0.0044)	0.902 (0.0042)	0.921 (0.0041)	0.867 (0.0044)
CNI	0.920 (0.0066)	0.920 (0.0064)	0.937 (0.0057)	0.891 (0.0077)
KS	0.927 (0.0046)	0.928 (0.0045)	0.943 (0.0039)	0.901 (0.0055)
Test Guil 2019 (Temporal transfer)				
None	0.904 (0.0037)	0.904 (0.0036)	0.917 (0.0034)	0.886 (0.0042)
CNI	0.923 (0.0093)	0.923 (0.0093)	0.933 (0.0079)	0.908 (0.0113)
KS	0.931 (0.0102)	0.931 (0.0102)	0.939 (0.0088)	0.919 (0.0120)
Test Gyrone 2018 (Spatial transfer)				
None	0.869 (0.0092)	0.871 (0.0089)	0.886 (0.0096)	0.847 (0.0080)
CNI	0.905 (0.0110)	0.906 (0.0107)	0.919 (0.0104)	0.885 (0.0111)
KS	0.905 (0.0094)	0.906 (0.0091)	0.919 (0.0087)	0.884 (0.0098)

5.4. Comparison with existing products

5.4.1. THEIA L2B snow product

The qualitative assessment in Section 5.3.2 shows that our models produce smoother maps that do not exhibit the geometric effect of the projection of MODIS NDSI in the SAR geometry. To evaluate our model prediction at a resolution in the same order of magnitude as the one of the Sentinel-1 image (5×20 m) we also evaluate our models on THEIA L2B Snow product (20×20 m).

We also measure the agreement between thresholded MODIS NDSI and THEIA L2B Snow to get the baseline performance, summarized in Table 7. MODIS NDSI and THEIA L2B Snow product exhibit a good agreement with an Overall F1-score above 0.88 over the Guil basin for 2018 and 2019 water year. There is less agreement over the Gyrone basin, likely due to a rougher terrain. Fig. 11 shows a snow cover map of the same day from MODIS NDSI and THEIA L2B Snow, we see that the small variations in snow cover distribution cannot be captured by MODIS NDSI coarse resolution.

Table 7 also shows model performance for different interpolations on THEIA L2B Snow labels. The interpolation also increases performance, with the model trained on KS pseudo-labels being the best model. However, the performance are lower than the one measured on MODIS NDSI and summarized in Table 6. The Overall F1-score loses 0.065 when evaluated on the Guil 2018. Fig. 11 shows that the classification errors are mostly present in the interface between Snow and NoSnow classes. The model does not manage to predict the granularity at the interface even if it predicts on a 5×20 m grid, because of the low-resolution training labels.

Even if these metrics are lower than the one computed on MODIS NDSI, the models applied only on SAR dual-pol amplitude images achieve Overall F1-score above 0.86 and are comparable to purely optical based snow products.

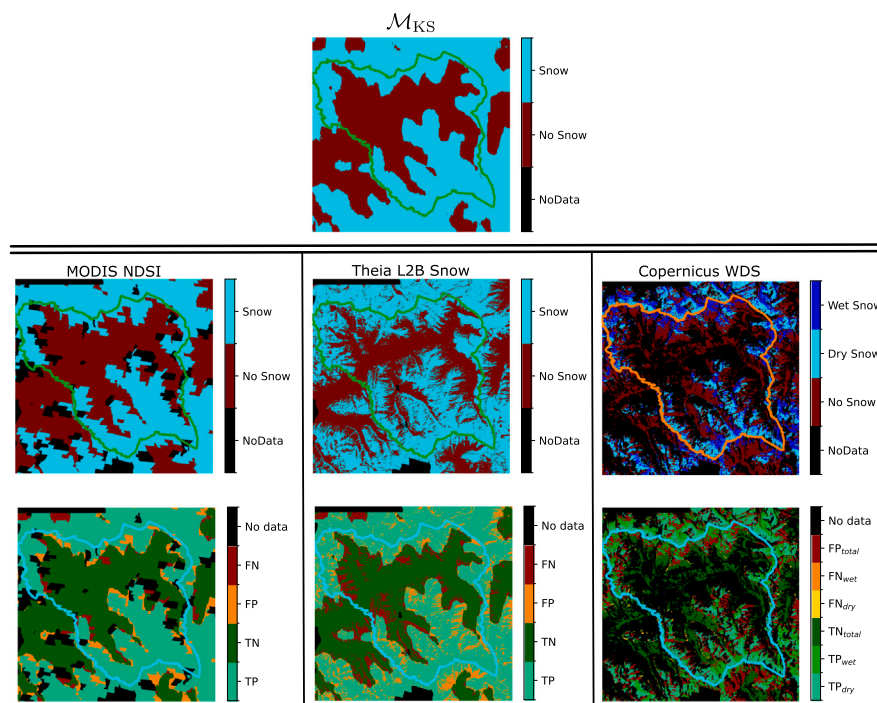


Fig. 11. March 31st, 2019 snow cover map prediction from $\mathcal{M}_{KS}$ (without domain transfer), with label maps from MODIS NDSI, THEIA L2B Snow and Copernicus WDS, as well as the corresponding label agreement maps. Copernicus WDS does not estimate snow cover in vegetated, water, or urban areas, nor in regions affected by SAR shadow, foreshortening, or layover. Greenish colors indicate correct predictions, while warm colors indicate model errors.

5.4.2. Copernicus wet/dry snow product

To assess the performance of our method on wet and dry snow independently, we also evaluate it on Copernicus WDS product. Since the models only predict binary snow cover, we compute only the recall for wet and dry snow classes. Table 8 shows recalls on both type of snow for models trained with our different interpolation methods. Label interpolation improves recall for both snow types. All models, except $\mathcal{M}_{KS}$ in the spatial transfer setting, achieve higher recall on dry snow than on wet snow. Temporal transfer does not significantly affect performance in either class, while spatial transfer leads to a drop in recall for both wet and dry snow. Fig. 11 also includes a label map from Copernicus WDS product, along with a label agreement map. Both types of snow are detected accurately and, as with previous comparisons using MODIS and THEIA L2B Snow products, most segmentation errors are located at the interface between snow-covered and snow-free areas.

6. Discussion

6.1. Selection of SAR channels

Quantitative and qualitative evaluation of the different channel sets shows that adding a reference image, as a concatenation or a ratio, always improves model performance. This allows the network to have information, in pixel values, relative to a snow-free state. However, the ratio removes information about elevation, which helps the network to predict less noisy snow cover maps and more accurate interface between snow covered and snow free regions as seen in Fig. 5. It seems that the reference image also contains topographical information that is useful for snow cover prediction, and using it for normalization removes this important part. Gallet et al. (2024) showed that the inclusion of topographical information (elevation and slope) for the classification of wet snow patches significantly improved performances. They used ratios between raw backscatter amplitude and reference images, which also improved performance from raw backscatter but did not try a direct concatenation between both. Daudt et al. (2023) also uses elevation maps as input to their RNN. We chose not to use

topographical information as we wanted our model to be able to predict snow only from SAR images, and prevent it to overfit on elevation information due to its strong correlation with snow presence. Even if we do not explicitly provide topographical information, the model is able to use the spatial features in the reference image to predict more accurate snow cover maps.

By using snow physics-aware channels to help the network, we have shown that keeping low-frequencies prevents the network to have topographical information and the noisy inputs prevent good snow cover segmentation for the network. As we said in Section 1, Nagler and Rott (2000) used spatial filtering methods to remove the noise in R_{wet} in order to apply a fixed threshold, however we expected the network to be able to handle this noise and act as an adaptive thresholder in order to detect wet and dry snow. We can note that the reference image is a temporally filtered image, and brings spatial information without taking the risk of removing possibly useful features by filtering the acquisition at time t . R_{wet} is a linear combination of channels set **C**: the additional information in channel set **D** is contained in R_{dry} , which has the purpose to help the network to detect dry snow. However, the only effect seems to make model predictions less stable. In Lievens et al. (2019), the ratio $\sigma_{VH}^0/\sigma_{VV}^0$ is computed to see the increase of σ_{VH}^0 due to dry snow accumulation. The ratio with σ_{VV}^0 is used to be independent from temporal variations such as vegetation growth, ground surface changes or snow conditions that impact both σ_{VV}^0 and σ_{VH}^0 . To retrieve snow height, this ratio is compared to the one of the previous day. The difference between those two terms is correlated to snow height variation between the two days. In our application, R_{dry} is computed from one reference image instead of using consecutive observations. The ratio is representative of the depolarization from a snow-free state, but the reference might be too far from observations to be relevant for the model.

6.2. Weak supervision through label interpolation

6.2.1. Data missingness in the NDSI signal

Cloud cover is the primary source of missing NDSI observations in the MODIS time series. Over the Guil basin during the 2018 water

Table 7

Mean performance and standard deviations of models trained with different interpolation methods, evaluated on the THEIA L2B snow product. These labels have a 20 m spatial resolution and are derived from Sentinel-2 and Landsat-8 data. For comparison, we also report the scores of MODIS-derived label maps in the first row for each domain. The best model scores are shown in bold.

Interpolation	Accuracy ↑	Overall F1 ↑	F1-Snow ↑	F1-No Snow ↑
Test Guil 2018 (Same domain)				
MODIS NDSI	0.896 (0.0000)	0.895 (0.0000)	0.920 (0.0000)	0.850 (0.0000)
None	0.813 (0.0037)	0.815 (0.0033)	0.847 (0.0041)	0.759 (0.0021)
CNI	0.843 (0.0061)	0.843 (0.0053)	0.876 (0.0069)	0.786 (0.0025)
KS	0.864 (0.0031)	0.863 (0.0031)	0.895 (0.0025)	0.808 (0.0053)
Test Guil 2019 (Temporal transfer)				
MODIS NDSI	0.882 (0.0000)	0.881 (0.0000)	0.888 (0.0000)	0.875 (0.0000)
None	0.813 (0.0034)	0.813 (0.0033)	0.825 (0.0040)	0.799 (0.0031)
CNI	0.821 (0.0037)	0.820 (0.0040)	0.842 (0.0042)	0.795 (0.0073)
KS	0.862 (0.0038)	0.861 (0.0038)	0.875 (0.0034)	0.845 (0.0047)
Test Gironde 2018 (Spatial transfer)				
MODIS NDSI	0.821 (0.0000)	0.816 (0.0000)	0.847 (0.0000)	0.782 (0.0000)
None	0.715 (0.0051)	0.713 (0.0049)	0.744 (0.0085)	0.677 (0.0082)
CNI	0.787 (0.0061)	0.786 (0.0055)	0.810 (0.0088)	0.758 (0.0030)
KS	0.781 (0.0051)	0.779 (0.0049)	0.804 (0.0061)	0.751 (0.0057)

year, approximately 52% of observations are missing. Moreover, snow events tend to occur under cloud cover, which introduces a sampling bias in the observed NDSI time series. Fig. 12 shows the mean fraction of missing data as a function of altitude in the Guil basin. A weak negative correlation is observed between altitude and data missingness ($r = -0.3$), suggesting that lower elevation areas tend to experience slightly more cloud cover, which we can also see in Fig. 13, which shows the spatial distribution of missing observations in our study area.

Temporal interpolation with CNI and KS is used to increase the amount of available supervision during training.

6.2.2. Effect of 0.4 fixed NDSI threshold for generating labels

Our method generates labels from NDSI using a fixed threshold of 0.4 established by Hall et al. (1995), corresponding to at least 50% snow coverage within a pixel. This value is standard in the literature and is used as an initial threshold in MOD10A1 Snow Cover and THEIA L2B Snow products. We acknowledge that this threshold may be suboptimal for certain terrain types. Indeed, Härer et al. (2018) demonstrated using Otsu's clustering-based segmentation that local-scale errors can reach 24.1% on 30 m Landsat products in Vernagtferner in Austria using this fixed value. However, they also showed that the 0.4 threshold remains appropriate at resolutions no better than 500 m. Since MODIS NDSI data is acquired at a resolution of 500 m, the fixed threshold is suitable for our framework. Forest areas or dirty snow typically require lower NDSI thresholds, which could lead to underestimation of snow cover in these regions. To assess the impact of this threshold choice, we analyzed the NDSI distribution in our training set (September 2018 to June 2019). The distribution exhibits two distinct modes: applying Otsu's method to maximize the inter-class variance yields a threshold of 0.244. However, these two modes do not necessarily

Table 8

Mean recall and standard deviation for wet and dry snow classes, evaluated with Copernicus WDS data.

Interpolation	Recall Wet ↑		Recall Dry ↑	
Test Guil 2018 (Same domain)				
None	0.931	(0.0161)	0.963	(0.0061)
CNI	0.954	(0.0108)	0.971	(0.0041)
KS	0.960	(0.0072)	0.976	(0.0012)
Test Guil 2019 (Temporal transfer)				
None	0.921	(0.0223)	0.974	(0.0043)
CNI	0.955	(0.0090)	0.979	(0.0032)
KS	0.956	(0.0074)	0.975	(0.0051)
Test Gironde 2018 (Spatial transfer)				
None	0.866	(0.0355)	0.879	(0.0207)
CNI	0.919	(0.0195)	0.927	(0.0124)
KS	0.936	(0.0082)	0.920	(0.0111)

correspond to *Snow* and *NoSnow* classes but rather the two dominant spectral behaviors in the dataset. 5.8% of the training labels are in the [0.244;0.4] interval. We show in Fig. 14 their spatial distribution for two different dates. Pixels in the interval are predominantly located near the interface between *Snow* and *NoSnow* classes, showing that any underestimation should primarily affect class boundaries rather than the core snow-covered areas. We emphasize that the 0.4 threshold is used only to generate training labels, not for final snow detection. The CNN learns to map SAR features to these NDSI-derived labels and may implicitly compensate for terrain-specific biases by learning more robust representations. An analysis of the ratio of false positives to false negatives showed that our models tend to overestimate rather than underestimate snow cover, regardless of whether raw or interpolated labels are used and which domain is tested. This suggests that our deep learning framework effectively overcomes the conservative bias introduced by the 0.4 threshold. However, because our primary validation uses labels derived from this same 0.4 threshold, the evaluation may appear circular. We therefore also validated our method against the independent THEIA L2B Snow product, which relies on the Let-It-Snow algorithm. This algorithm performs two consecutive passes with different NDSI thresholds (0.4 for pass 1, 0.15 for pass 2) and includes additional spectral tests on red reflectance. Our method achieves an Overall F1-Score of 0.86 against this independent reference, confirming that our framework is not limited by the fixed threshold used for label generation.

6.2.3. Bias and variance tradeoff of the Kalman smoother

The KS not only interpolates missing NDSI values but also estimates NDSI values at observed dates. In this subsection, we assess the bias and variance tradeoff of the KS. Denoting $\bar{\epsilon}$ the mean bias and σ_{ϵ} the standard deviation of the estimation error between the smoothed NDSI and the MODIS observations, the mean prediction bias is close to zero ($\bar{\epsilon} = -0.003$), so the smoother does not introduce a significant amount of systematic error when predicting NDSI, which ranges from -1 to 1 . In comparison, the purely random component is an order of magnitude higher ($\sigma_{\epsilon} = 0.042$). We further separated the spatial and temporal components by averaging over each pixel's time series and over each NDSI map respectively. Fig. 15 shows the spatial distribution of the estimation error. The spatial bias is overall very low, though low-altitude pixels tend to be slightly overestimated while higher elevations exhibit larger variance in bias. Fig. 16 shows the temporal evolution of the spatially-averaged error: no seasonal trend is observed in either the RMSE or the bias, confirming that the smoother does not introduce a significant amount of systematic seasonal distortion. The bias averaged per map is however approximately 10 times higher than the global mean bias, suggesting that while there is no persistent bias, individual acquisitions can be more affected by regularization, particularly during rapid snow cover transitions.

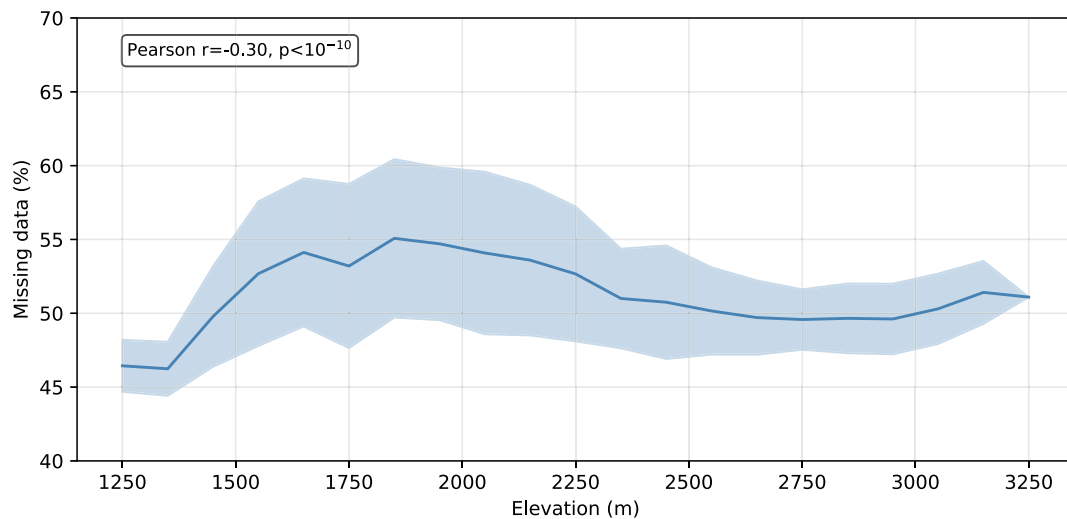


Fig. 12. Mean fraction of missing MODIS NDSI observations per 50 m elevation bin over the Guil basin (September 2018 to June 2019). The shaded area represents ± 1 standard deviation. The Pearson correlation coefficient between elevation and data missingness is $r = -0.3$ (p – value $< 10^{-10}$).

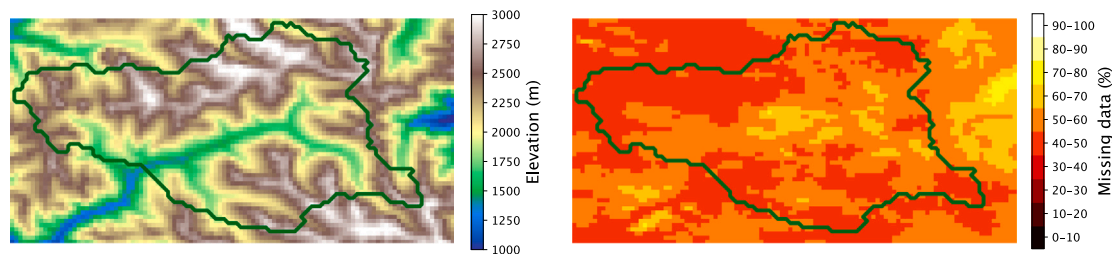


Fig. 13. Spatial distribution of the fraction of missing MODIS NDSI observations over the Guil basin (green line) from September 2018 to May 2019 (right), and the associated elevation map (left).

6.2.4. Impact of the interpolation methods on the model performance

Increasing the level of supervision through label interpolation, whether using CNI or KS, improved model performance for every task. This improvement was most apparent in qualitative evaluation, where label interpolation produced to smoother predicted snow cover maps. This increased smoothness can be attributed to the gap-filling applied to the training labels: even though the interpolation is purely temporal, spatially close pixels often exhibit strong correlation, resulting in smoother training images. As a result, the network is exposed to more spatially coherent supervision. However, this increased smoothness can also reduce the model ability to capture sharp transitions between snow-covered and snow-free areas, slightly affecting the spatial precision of the predictions. This is especially relevant in SAR imagery, where sharp transitions are expected due to geometric distortions such as layover or foreshortening.

6.2.5. Setting η value in the Kalman smoother

Increasing the amount of supervision with interpolation helps the model, but the KS provides a continuum of interpolated signals through its parametrization with the regularization parameter η , ranging from closely following observations (similar to linear interpolation between observations) to producing highly smoothed signals where temporal trends dominate and short-term fluctuations are suppressed. Since these time series are used to assign labels to our pixels, excessive regularization introduces unintended label noise by overly damping oscillations around the classification threshold or by removing potential short-time events. In the opposite case, no regularization keeps the original label noise. In the absence of ground-truth, the optimal level of regularization must be determined empirically. As shown Fig. 7, η values smaller than 0.3 produce almost identical time series, leading only to few decision changes in the pseudo-labels. Indeed linear interpolation

is a limit case of the Kalman smoothing for small η . However, as shown in Fig. 9, these small changes still lead to an increase in model performance for η ranging from $\eta = 0.001$ to $\eta = 0.1$. These changes are concentrated near the snow line, where pixels with NDSI values close to the threshold are most sensitive to regularization, as shown in Fig. 8. For larger η values, the performance becomes unstable due to the over smoothing of the NDSI time series. Even if other η values could have lead to higher performance, we choose to keep $\eta = 0.3$ to ensure the robustness of the model. In our setting, we fixed $v_{\text{mod}} = 0.001$ and tuned η over $[10^{-3}; 10^2]$, effectively treating η as a measure of our confidence in the observations relative to the model. If one had more information on the quality of their observations, v_{obs} could be fixed instead and tuning η would amount to tuning the model noise, *i.e.* the expected NDSI step between two consecutive dates.

6.2.6. Transfer to other domains

In temporal transfer, all models yield better performance than when tested on the same domain as training, which may seem counter-intuitive. This is explained by the composition of the Guil 2019 test set, which spans the time series from September 1st to June 1st, whereas the Guil 2018 test set covers only a limited subset of the same hydrological period from the previous year (Fig. 4). Thus, the 2019 set includes more observations taken during early autumn and mid-winter, when one class tends to dominate. As illustrated in Fig. 5, models perform well when predicting large snow covered or snow free areas, while fine-grained segmentation near class boundaries remains more challenging.

The performance gap between predictions in the Guil and Gyrone basins may come from the limited size and diversity of our dataset, which consists of the time series of a single basin, over one water year. While it is enough to train models that perform well on the

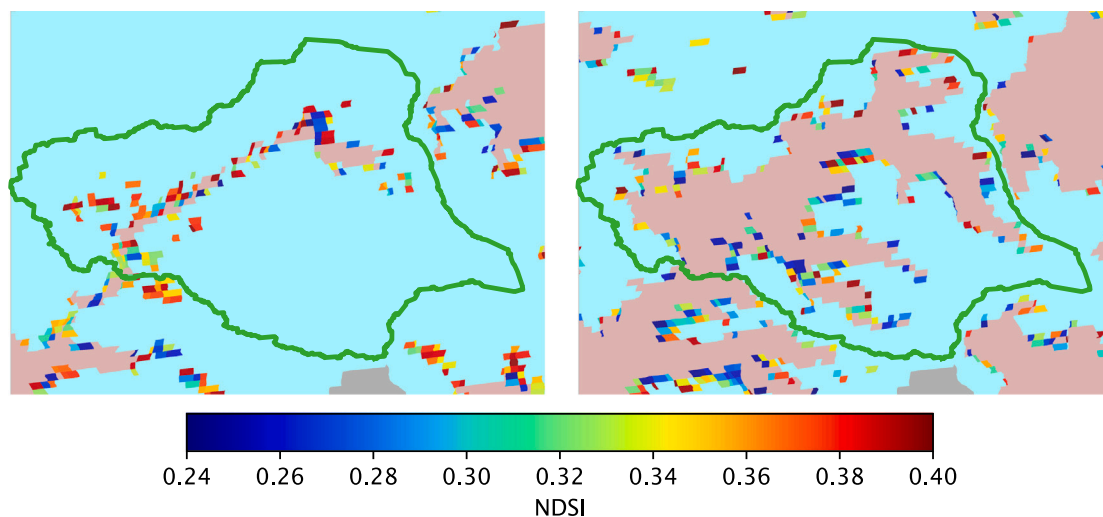


Fig. 14. Spatial distribution of NDSI values in the interval $[0.244; 0.4]$, illustrated for two SAR acquisition dates: December 12th 2018 (left) and March 18th 2019 (right). The background shows the binary labels, interpolated with KS with $\eta = 0.3$ to show the whole classes interface ($\square$: Snow, $\square$: No Snow, $\square$: NoData). The green line shows the Guil basin.

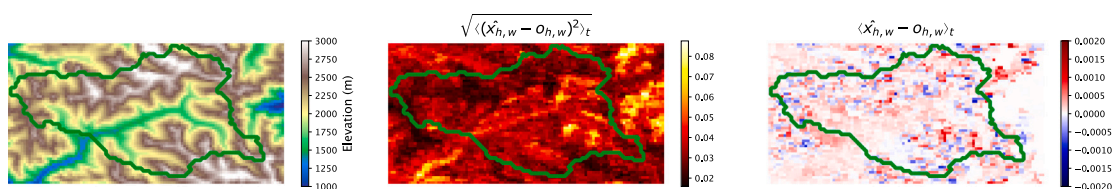


Fig. 15. Pixel-wise temporal reconstruction errors of a Kalman smoother ($\eta = 0.3$) predicting NDSI over the Guil basin in the MODIS geometry, where $\hat{x}_{h,w}$ denotes the smoothed NDSI predictions and $o_{h,w}$ the raw MODIS NDSI observations. Elevation map shown for spatial reference (left), temporal mean RMSE (center), and temporal mean bias (right). The green contour delineates the Guil basin.

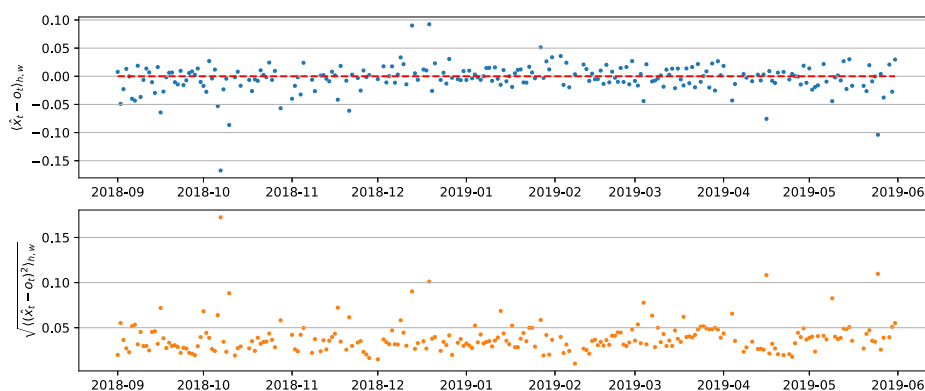


Fig. 16. Temporal evolution of the spatially averaged bias (top) and RMSE (bottom) of a Kalman smoother ($\eta = 0.3$) predicting NDSI over the Guil basin in the MODIS geometry, where $\hat{x}_t$ denotes the smoothed NDSI predictions and o_t the raw MODIS NDSI observations. The dashed red line indicates zero bias.

same domain or under small temporal shifts, the SAR features may not capture enough variability to allow robust generalization to new regions. Our interpolation method does not tackle this issue as it can only increase the amount of optical labels, not diversify the SAR features.

6.3. Comparison with existing products

We validated our method using the THEIA L2B Snow product, which is derived from Sentinel-2 and Landsat-8 optical imagery and maps snow cover. While it provides higher spatial resolution than MODIS, it is only available every 5 to 10 days. All dates with a snow cover map from THEIA L2B Snow were included in the Guil 2018 test set,

so that the dates used to test models remain the same across datasets. The results listed in Table 7 confirm the gain of using pseudo-label obtained by temporal interpolation of the snow product during training. However, the comparison with the Overall F1-score evaluated on the MODIS NDSI product listed in Table 6, shows that all the models are lower when evaluated on the THEIA L2B Snow product than when evaluated on the MODIS NDSI.

To evaluate how this loss can be linked to the use of coarser resolution label during training, we performed the direct comparison between the thresholded MODIS NDSI and the THEIA L2B Snow product. Although the agreement between the two products is high, the finer resolution of the THEIA L2B Snow product reveals small-scale features that are not captured in the coarser resolution MODIS NDSI. Thus, most disagreements are located at the boundaries between the Snow and No

Snow classes, particularly at mid-altitudes, where rapid spatial changes in snow cover require higher-resolution observations. This is especially true for the Gyrone basin where the disagreement between the MODIS NDSI and the THEIA L2B Snow product was the highest.

The model trained using the Kalman smoothed pseudo-label achieves performance close to the agreement between the two products, showing its ability to predict snow cover from SAR dual-polarization amplitude images. Unlike MODIS NDSI, it produces spatially smooth snow cover maps, showing its capacity to build on small scale SAR features. However, classification errors are still mainly located at class boundaries, as shown in Fig. 11. In these transition areas, small-scale patterns are often missed, likely due to the coarse pseudo-labels derived from MODIS NDSI used during training.

To ensure the deep-learning models do not make errors on a single type of snow, we compared the estimated snow cover maps to the Copernicus WDS product, which distinguishes between wet and dry snow. The comparison showed that both snow classes were correctly detected, but with more errors on the *Wet Snow* class. This is somewhat counterintuitive, since dry snow is largely transparent to SAR signals.

However, Blöschl et al. (1990) showed that in a basin, the snow-pack at the lowest altitudes is saturated with liquid water and runoff-producing. The model lower recall on the *Wet Snow* class could be explained by its spatial repartition and the model difficulties to segment accurately *Snow* and *No Snow* classes boundaries, as shown in Fig. 11.

The small proportion of *Wet Snow* in the training set could also explain the model difficulties to segment this class. In the MODIS NDSI product from the 2019 Guil test, we observed a count of *Dry Snow* pixels twice higher than the count of *Wet Snow* pixels among the pixels classified as *Snow*. Because Sentinel-1 and Sentinel-2 have regular revisit time, we can assume this proportion is representative of the one in the Guil 2018 train set. An under-representation of the *Wet Snow* class in the positive examples given to the model during training could also explain why its associated recall is lower.

7. Conclusion

In this study, we proposed a method to predict binary snow cover maps from Sentinel-1 SAR backscatter amplitude. Training labels were generated in an unsupervised manner by thresholding MODIS NDSI time series from the MOD10A1 dataset. A U-Net architecture was trained for binary semantic segmentation using a standard cross-entropy loss.

We showed that the most effective SAR input configuration is the concatenation of co- and cross-polarized backscatter amplitude channels with reference images derived from snow-free acquisitions. More heavily processed, snow-specific SAR channels did not improve performance, likely due to the loss of topographic information that the network otherwise learns to exploit.

We proposed to generate pseudo-labels derived from MODIS NDSI time series interpolation to compensate for missing information due to cloud cover, and we investigated whether this can help the model to predict better snow cover maps. Two temporal interpolation methods were considered: Closest Neighbours Interpolation, and Kalman smoothing. The level of regularization in the KS was found to be critical: moderate values improved performance by balancing temporal consistency and fidelity to observations, whereas excessive smoothing led to information loss. We showed that models trained with interpolated labels achieved better accuracy and produced less noisy predictions. Under temporal domain shift, the models were able to generalize better when trained with pseudo-labels. However, spatial generalization remains a challenge, as the training set may lack diversity in SAR features to capture the variability across different basins. Increasing label density cannot fully address this limitation. Satellite time series generally involve a trade-off between spatial and temporal resolution. Because of the fast dynamics of snow cover, we chose to use

MODIS data, acquired every day, to ensure accurate temporal monitoring and reliable pseudo-labels despite a spatial resolution more than ten times coarser than the Sentinel-1 SAR images. The use of pseudo-labels helps to produce smoother snow maps but did not overcome the low resolution of the MODIS NDSI pseudo-labels.

These results highlight that snow cover maps can be estimated solely from SAR dual-pol amplitude images by leveraging the ability of optical data to jointly detect both dry and wet snow to supervise model training, and that adding pseudo-labels with temporal interpolation improves prediction in cloud-prone environments as well as the capacity of temporal generalization. Beyond the specific case of snow, our novel strategy illustrates a more general weakly supervised learning framework, where optical observations can be exploited to train SAR-based models. Our approach relies on comparing SAR acquisitions to a snow-free reference baseline, a principle that extends naturally to other environmental monitoring tasks. This reference-based comparison connects to the broader field of change detection, where multisource approaches combining optical and SAR data have shown particular promise (Jiang et al., 2024; Gong et al., 2024). Vision foundation models have also been recently adapted to such remote sensing change detection tasks (Jiang et al., 2025). For applications where changes occur at a slower rate than snow dynamics, our methodology could enable the use of optical data at finer spatial resolution with sparser temporal sampling. Rainforest deforestation monitoring is a natural candidate: it is typically based on spectral indices from optical observations, but hindered by persistent cloud cover in tropical regions. Our weakly supervised framework, which transfers optical-derived supervision to SAR-based inference, could help address this limitation.

CRedit authorship contribution statement

Swann Briand: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Flora Weissgerber:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Sylvain Lobry:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Jérôme Idier:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the support of the ONERA, the French Aerospace Lab and the Région Pays de la Loire.

Data availability

The dataset is made publicly available at <https://doi.org/10.57745/IMTSFL> (Briand et al., 2025).

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